

Toward Robot-Assisted Transcatheter Laser-based Cutting of Calcified Heart Valve Tissue

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INTRODUCTION

Due to the difficulty of mastering the complex manipulations needed to control catheters, robotic platforms are being developed for many procedures, such as valve repair and replacement [1], treatment of arrhythmia [2], and coronary interventions [3]. Improving catheter navigation precision through robot-assisted steering allows for precise maneuvering to target locations within the heart and broadens accessibility to reduced-risk transcatheter approaches by lowering the learning curve.

Challenges remain, however, in developing catheter-delivered tools that can replicate critical tasks done in open surgery. For example, while slicing or removing the aortic valve leaflets prior to transcatheter valve replacement is desired to avoid occlusion of the coronary arteries, transcatheter cutting of calcified leaflet tissue is an unsolved problem. Current approaches, which use electrosurgery, cannot penetrate through calcifications [4]. The hardest part of the procedure is the initial penetration or traversal through the base of the coronary leaflets, which is required to begin cutting them along their midline (see Fig. 1). While an alternative strategy using a cutting blade has been proposed, large forces are needed to penetrate heavily calcified leaflets [5].

In this paper, we investigate the use of laser-based tools as an alternative approach to transcatheter cutting of calcified tissue. A system consisting of a robotic catheter and steerable balloon cardioscope [6] (Fig. 2A) can be maneuvered to either coronary leaflet to perform traversal, replacing the tool positioning step shown in Fig. 1A with robotic-assisted guidance of positioning via direct intracardiac visualization of tissue. We conducted in vitro experiments to compare laser traversal of human leaflets removed during surgical aortic valve replacement (Fig. 2B) with the established electrosurgical and mechanical approaches (cutting tools depicted in Fig. 2C).

For laser-based cutting, we used a thulium fiber laser (TFL), which is effective in urological lithotripsy [8]. Experiments on phantom leaflet calcifications were used to determine laser parameters. These were followed by comparison testing on human leaflets, evaluating traversal success, time, applied force, and hole size.

MATERIALS AND METHODS

A test system (Fig. 3A) was designed to compare traversal of leaflets with electrosurgical (360 μ m wire mimicking the standard approach [4]), laser (365 μ m optical fiber powered via TFL (Quanta Systems)), and mechanical (413 μ m needle) cutting tools (Fig. 2C). The

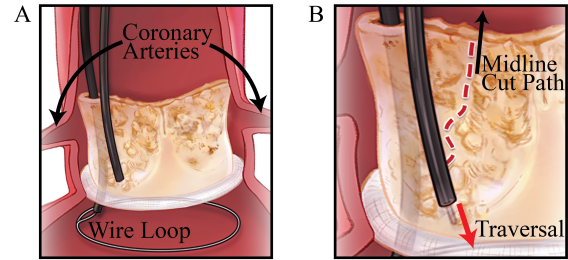


Fig. 1 Standard electrosurgical cutting approach. A. Manually steered catheter is positioned at the base of the calcified aortic valve leaflet cusp. B. The tool is traversed through the leaflet and snared by the loop for cutting the leaflet down its midline.

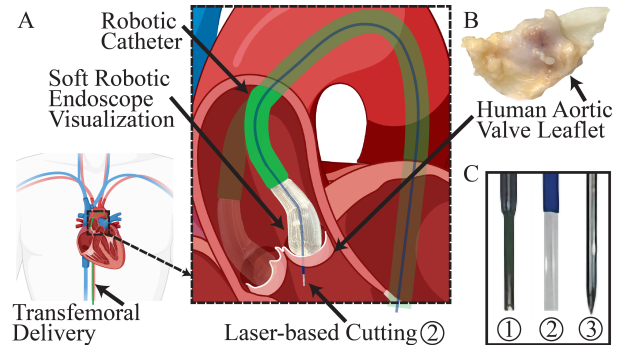


Fig. 2 Envisioned laser-based cutting approach. A. Robotic delivery of laser cutting tool with direct cardioscopic visualization. B. Calcified human aortic leaflet. C. Cutting tools to compare: (1) Electrosurgical, (2) laser, and (3) mechanical [7].

system consists of a vertically translating stage to position a tool against a horizontally mounted leaflet with an adjustable applied contact force. Traversal times were measured with sensors monitoring the start/stop (energy initiation/traversal completion) of each test. Pin gages were used to determine the traversal hole diameter.

Phantom calcified leaflets were fabricated as square blocks of Begostone Plus (used to model renal calculi [9]) mounted as in Fig. 3B. These experiments compared the tools and determined the optimal TFL parameters (frequency and power) for efficient laser traversal. With an initial applied force of 0.5N, electrosurgical and laser energy was applied (max 60s) and mechanical tests increased force incrementally until traversal or tool buckling (≈ 20 N). During electrosurgical tests, power level began at 50W (matching standard of [4]) and was incremented up to 300W. Similarly, contact force was increased to 1N after initial trials at 0.5N. Laser traversal was performed at frequencies of {10,20,50,100}Hz and power levels of {10,20,30,40,50}W.

These preliminary phantom tests were important to

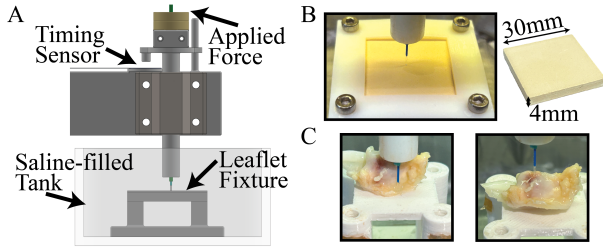


Fig. 3 Traversal apparatus. A. Test components. B. Calcified leaflet phantom in fixture and block size. C. Non-calcified (left) and calcified (right) regions of human leaflets in the fixture (modified without top cover of fixture to allow visualization).

inform our human leaflet experiments because we were limited to a sample size of 5 native leaflets. Using the fixture of Fig. 3C, traversal was attempted in non-calcified and calcified regions of each leaflet with the thickest calcifications averaging 5mm. With the exception of laser traversals utilizing only the optimal parameters, the testing parameters were the same as in phantom experiments.

RESULTS

A. Phantom Calcified Leaflet Traversal Tests: Despite increasing the applied force to 1N and using power levels up to 300W, electrosurgical traversal was never successful. Likewise, mechanical traversal was unsuccessful with the needle failing due to buckling. Laser traversal was always successful with 20Hz exhibiting the most consistently low traversal times (Fig. 4). Between 30-50W traversal time averaged 0.8 ± 0.4 s independent of frequency. Furthermore, hole diameter variation in this range is modest ($513 \pm 102 \mu\text{m}$). Across frequencies, 40W produced the smallest holes ($486 \pm 84 \mu\text{m}$) and most consistent traversal times (0.6 ± 0.2 s). For these reasons, laser parameters of 40W, 20Hz were selected for native leaflet testing.

B. Native Leaflet Traversal Tests: In non-calcified regions, all traversal attempts succeeded and penetrated in under 1s, requiring only 50W at 0.5N force for electrosurgery, 0.5N for laser, and 0.7 ± 0.3 N for mechanical.

In calcified regions (Table I), electrosurgery was unsuccessful regardless of power or contact force. Mechanical traversal was more successful, failing in only 1/5 leaflets and making holes that matched the size of the needle. Laser traversal was always successful and, on average, created hole diameters of $456 \mu\text{m}$ in under 1s.

DISCUSSION

Our results confirm clinical insight that electrosurgical tools cannot effectively traverse calcifications. Although mechanical cutting was able to penetrate calcified leaflets, the higher forces required (max 10N) may be difficult to achieve without a dedicated mechanism, potentially complicating integration with robotic delivery.

The laser was highly effective in traversing calcifications, requiring minimal contact forces and generating hole sizes less than $100 \mu\text{m}$ larger than the diameter of the tool in native leaflets. We consider smaller hole sizes safer, as larger holes could indicate an increased risk of embolism formation. Though more work is needed to investigate the effects of debris particles generated from all cutting methods, a similar laser produced mostly particles that were $< 6 \mu\text{m}$ (85%) and few $> 300 \mu\text{m}$ (0.1%)

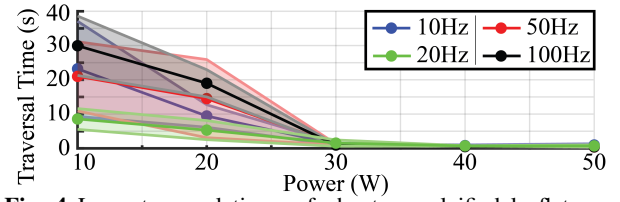


Fig. 4 Laser traversal times of phantom calcified leaflets as functions of power for each frequency. The solid line is the mean and the shaded area is the standard deviation.

TABLE I Calcified Traversal Results.

		Laser	Electrosurgery	Mechanical
Phantom	Success (n=5)	100%	0%	0%
	Force (N)	0.5	0.5, 1	0.5-20
	Time (s)	0.7 ± 0.05	-	-
	Diameter (μm)	496 ± 23	-	-
Native	Success (n=5)	100%	0%	80%
	Force (N)	0.5	0.5, 1	5.8 ± 2.7
	Time (s)	0.8 ± 0.7	-	-
	Diameter (μm)	456 ± 34	-	413

when cutting calcified aortic leaflets, suggesting TFLs are generally safe and suitable for cardiac applications [10].

Because of the prevalence of calcification in valve replacement patients, the ability of the laser to consistently penetrate all tissue regions positions it as a compelling alternative to overcome the limitations of standard cutting tools. Further, robotic delivery of laser-based tools (optical fibers can easily deploy through the working channel of the proposed robot of Fig. 2A) with direct tool-tissue contact visualization is a promising alternative to standard manual delivery and imaging methods. The next step is to integrate catheter-delivered tools with robotic platforms to enable all the capabilities of open surgery to be deployed through safer transcatheter approaches.

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